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Review

The regulation strategy for releasing residual stress in ceramic-metal brazed joints

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Abstract

Ceramics and ceramic-based composites have been widely used in electronic information and communication, chemical industry, instrument manufacturing and other fields. However, due to the brittleness and processability of ceramic materials, ceramics need to be connected with metal materials to meet practical applications. Among various bonding methods, brazing is widely used for joining ceramics and metals. However, due to the difference in the properties of ceramics and metals, especially the coefficient of thermal expansion, excessive residual stresses are generated in the joint subject to deformation during the cooling process of the brazed joint. High residual stresses can seriously damage the joint properties. To relieve residual stresses, this review has summarized

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three residual stress relief methods: particle reinforcement, interlayer assisted control, and surface structure design. These methods can effectively relieve stress concentration in joints, improve joint plasticity, and reduce the ceramic-metal discrepancy, thus leading to the enhanced performances. Finally, the future opportunities and challenges of regulation strategy for releasing residual stress are also presented. Overall, it is expected that these stress relief strategies will provide some new understanding of the wide range of applications for ceramic-to-metal joints.

Keywords

Particle reinforced brazing alloy; Interlayer assisted control; Surface structure design; Residual stress; Ceramic and metal brazed joint

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1. Introduction

Due to the advantages of high strength, low density, ablation resistance, and special photoelectric properties, as well as good thermal conductivity and corrosion resistance, ceramics and ceramic matrix composites have a wide range of application prospects in aerospace, electronic information and communication, instrument manufacturing and other fields [1], [2], [3]. Ceramics and ceramic matrix composites play an irreplaceable role in the manufacture of functional components, anti-corrosion and heat protection systems and lightweight structures. For example, aerospace engine nozzles use C/SiC composites [4], [5], and wideband materials represented by SiC and GaN are the representatives of the third generation of semiconductors [6], [7]. However, due to the high brittleness, poor processing and high cost of ceramic materials, it is difficult to make large-size and complex-shape components in industrial production. Therefore, it needs to be combined with other materials to form composite components for industrial

6. Conclusions and perspective

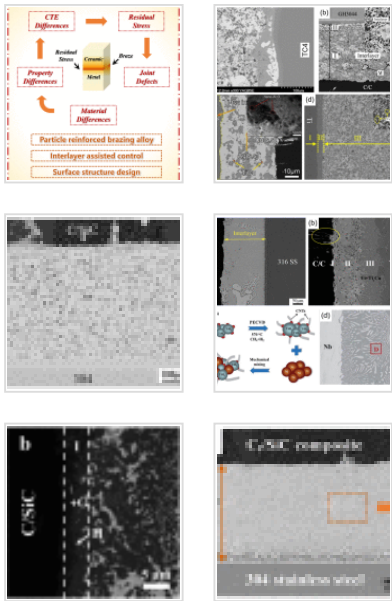
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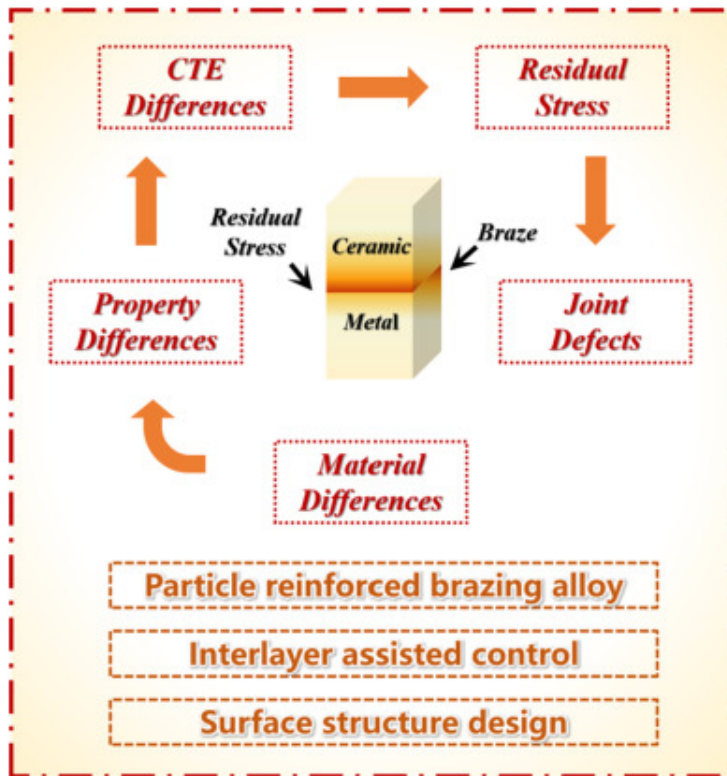
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applications. Metal materials with good plasticity and toughness and good processing properties are an essential part of industrial production. Metal materials can obtain different properties through material design to form alloys, such as titanium alloy [8], [9], [10] with high strength and good heat resistance, and shape memory alloy [11], [12] with shape memory effect. These properties can be combined with ceramic materials to form a complementary material for a wide range of applications in aerospace, energy, and medical fields. Today, ceramic-metal composite components play a key role in various fields. In aerospace, ceramic matrix composites and metal composites are used in a number of important components [13], such as engines, protection systems. In medicine, titanium alloys and bioceramic composites [14] also have important applications in dentistry and orthopedics. In electronic components, semiconductor ceramics and metals can form many important components [15]. Consequently, ceramic-metal composite components have been widely used in our life, and have a very important application value.

The application of ceramic and metal composite components requires the connection of ceramic and metal. Considering the characteristics of dissimilar material properties between ceramics and metals, fusion welding is not suitable for the connection of the two. In addition to the brittleness of ceramics, pressure welding is not suitable for joining ceramics to metals, either. In contrast, brazing is the most effective method, which is simple, efficient, and simultaneously suitable for joining dissimilar materials. To meet the requirements of different kinds of ceramics and alloys, many brazing systems [16], [17], [18] such as AgCuTi, TiZrNiCu, etc. have been developed and applied. Brazing has been very widely used in the connection of ceramics to metals, but there are still problems to be solved between the connection of the two [19], [20]. Due to the difference between the covalent bonding in ceramics and the metallic bonding in metals, there are differences in the properties of the two. In particular, differences in thermal expansion coefficient (CTE) and modulus of elasticity can generate excessive residual stresses during cooling due to deformation limitations. The high residual stresses directly weaken the bond strength of the ceramic-to-metal brazed joint and even lead to microcracks and cracks. As the size of

the structure and the conditions of use continue to increase, the requirements for joints and reliability increase further, but micro-cracks and cracks in large-size joints will become unstable factors. Effective relief of residual stresses in joints for better strength and serviceability has been the focus of research in ceramic-to-metal brazing. Among the contemporary research methods, there are three main methods to relieve residual stresses, including particle-reinforced composite brazing, interlayer assisted control, and surface structure design methods. In the method of particle reinforced brazing alloy, ceramic particles, carbon materials, negative expansion materials, and metal particles can bear the load, restrain the deformation of the substrate, regulate the solder properties and relieve residual stresses. The method of interlayer assisted control is to make the joint distribution more stable and with fewer welding defects through the interlayer structure between two pieces of brazing alloy, which can improve the joint plasticity and relieve the stress. In addition, there is a high-stress concentration at the junction of ceramics and metals due to the abrupt change in properties at their interface. The surface structure design of ceramics or its composite materials through perforation and etching can enhance the nailing effect of joints and realize the gradient shift of joint properties, thus relieving the stress concentration in joints. In summary, all of these methods can effectively relieve stress, ensure the quality of brazed joints, achieve reliable connections between ceramic and metal joints, and better serve a range of industrial manufacturing processes such as aerospace, communications, and energy.

In this review, we focus on the causes and characterization methods of residual stresses in ceramic-metal brazed joints and summarize the theoretical and experimental progress in recent years for the mitigation of residual stresses. As shown in [Fig. 1](#), the current stress relief methods are discussed primarily. Finally, we also discuss the challenges and strategic directions for future research on ceramic-metal brazing stress relief methods.



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Fig. 1. The schematic illustration of the modification strategy for ceramic-metal residual stress.

2. Generation and detection of residual stress in joints

To relieve residual stresses in ceramic and metal brazed joints, it is essential to understand the causes of residual stresses in the joint. This review summarizes the specific causes of stresses in brazed joints, and presents the main current methods for detecting residual stresses in joints to better understand the problems that exist in ceramic-to-metal brazing.

2.1. Causes of residual stresses in ceramics and metals

It is well known that materials will expand or contract as the temperature rises or falls. For the vast majority of materials, an increase in temperature results in an expansion of the material, and a decrease in temperature results in a decrease in volume. Of course, there are some materials whose volume remains almost the same or even decreases as the temperature increases, and these materials are called zero expansion materials or negative expansion materials [21]. To measure the change in volume of a material with temperature, we use the coefficient of thermal expansion to express their thermal expansion capacity. Many factors influence the coefficient of thermal expansion [21], [22], [23]. The composition of the material, crystal arrangement, bond energy strength, and phase composition all affect the coefficient of thermal expansion.

Ceramics are mainly composed of covalent bonds with strong bonding energy, and its strong bonding energy will give it a low coefficient of thermal expansion. Metals are mainly composed of metallic bonds with relatively weak bonding energy compared to ceramics, and therefore the vast majority of metals have relatively high coefficients of thermal expansion. Therefore, the connection of ceramics and metals can suffer from a mismatch of thermal expansion coefficients. In the process of heating or cooling, the ceramic and metal will generate thermal deformation with the temperature change, and under constraint, the thermal deformation will act on the ceramic-metal composite member to generate thermal stress. When brazing heating is performed, the joint is usually free of thermal stress. No thermal stresses are generated in the joint during the brazing heating process. During cooling, ceramics and metals have different shrinkage capacities, and under constraint, thermal stresses are transformed into residual stresses that are distributed in the weak areas of the joint [24], [25], [26], [27].

In general, the residual stresses in the joint are divided into transverse residual stresses and longitudinal residual stresses. As the joint is mainly constrained by the pressure in the vertical direction of the ceramic-metal contact surface during the brazing process, The residual stresses in the joint are mainly longitudinal residual tensile stresses. A defect in a

joint occurs when the external stresses and residual stresses are superimposed so that the total stresses exceed the strength limit of the joint [20]. Therefore, reducing the residual stress in the joint can improve the serviceability of the joint.

2.2. Method of detecting residual stress in joints

The detection of residual stress is very important to understanding the quality of the joint. With the progress of testing techniques, more and more testing methods have been applied to the residual stress detection of joints. These include X-ray diffraction [28], [29], hole drilling [30], nano-indentation [31], contour [32], [89], simulation methods [33] and neutron diffraction method [91], [92], which are important to ensure the quality of ceramic-metal brazed joints.

The X-ray diffraction method is a non-destructive testing method. Since there is elastic deformation in the crystal when there is residual stress in the joint, X-ray is used to calculate the residual stress in the joint by measuring the atomic spacing. However, the lattice structure can sometimes have an impact on the test results. The $\sin 2\psi$ method and the $\cos \alpha$ method are mainly used among the current main measurement methods. The $\sin 2\psi$ method is the most commonly used testing method, and the $\cos \alpha$ method is suitable for surface stress testing of large steel parts, which has certain limitations. In general, X-ray diffraction methods can be inspected without damaging the joint, but they are subject to errors due to the crystal structure of the material.

The hole drilling method is a traditional method of residual stress testing. By hole drilling small holes in the joint, the residual stresses in the joint are directly released. The amount of change in residual strain in this area is used to calculate the residual stress value at the hole. This method is simple and efficient, but it has deviations in the accuracy of the measurement and can also cause damage to the joint. Opening holes is also a very difficult task for ceramics, so the hole drilling method is not applicable to ceramic-metal composite components.

The nano-indentation method uses a special pressure head to press into the tested area, calculates the indentation-related parameters from indentation photographs of the material, and finally calculates its residual stress by an empirical formula. The measurement of nano-indentation reaches the micron level in scale. It can be categorized as a nondestructive method due to the low damage to the joint. It provides accurate measurement results, but due to the small measurement area, residual stress analysis needs to be performed in numerous areas of the joint to ensure a grasp of the residual stresses in the overall joint.

The contour method is developed by Prime [89]. It analyzes the residual stresses in the joint by contour the deformation of the joint after cutting. A part is cut in half and the contours of the resulting new surface are measured to determine the displacement caused by residual stress relief. Based on Bueckner's superposition principle, this method gives the original residual stresses normal to the cutting plane. This method is very simple and can capture the overall stress changes in the joint. However, this method can also cause direct damage to the joint.

The simulation method generally refers to the finite element simulation method. The joint is modeled by finite element software, and after adding relevant parameters, the whole joint is divided into many elements or meshes, and the residual stresses in the joint are calculated under the set conditions. This method does not require direct manipulation on the joint. When the cell is sufficiently small, the true residual stresses in the joint can be effectively calculated. In addition, the finite element simulation can visualize the overall distribution of residual stresses in the joint, which can observe the stress changes and predict the properties of the joint. However, the finite element simulation is time-consuming and more complicated [90], while the relevant parameters need to be prepared and the results of the analysis are influenced by the model.

The neutron diffraction method needs to analyze the structure and magnetic structure of crystals by using the diffraction pattern formed when neutrons are scattered on the crystal surface. This method has a high resolution, large coverage, and penetration

capability several centimeters deep, especially for bulk crystals, composite materials and brazing parts [91], [92]. But it has a low measuring efficiency and high cost, which is not conducive to the measurement of residual stress.

It is clear from the above that the problem of residual stresses is central to the brazing of ceramic to metal connections. The presence of residual stress can have many negative effects on the mechanical properties, corrosion resistance and other properties of the joint. Residual stress relief of the joint is essential to achieve a high-quality connection between ceramic and metal. In order to achieve effective relief of residual stresses in joints, this review summed up the following three types of residual stress relief methods to better ensure the quality of joints.

3. Particle reinforced brazing alloy

The overall thermal expansion coefficient of the joint is reduced when the particulate phase is uniformly present in the brazing alloy. As shown in Eq. (1), the low expansion particle phase effectively reduces the overall thermal expansion coefficient of the joint.

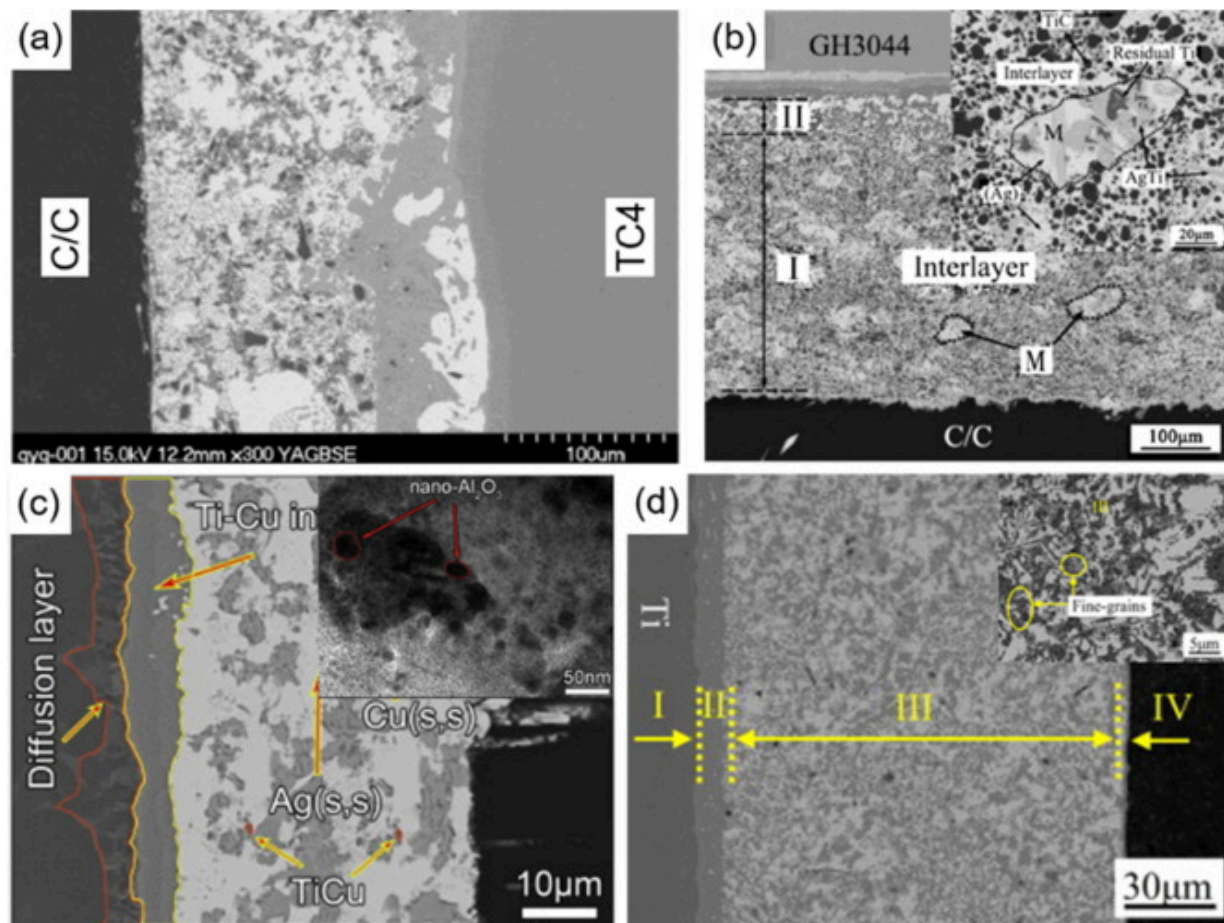
$$\alpha_m = \alpha_p V_p + \alpha_c V_c \quad (1)$$

where α is coefficient of thermal expansion, V is volume fraction, m is composite material, p is particle reinforced phase, and c is metal matrix.

The interface between the particles and the brazing alloy will share some of the stress and the stress concentration in the joint will be greatly relieved. The presence of the particulate phase prolongs the crack expansion path and effectively increases the joint ground bearing capacity [34], [35]. The current research on particle-reinforced phases focuses on ceramic particles, carbon materials, negative expansion materials and metal particles.

3.1. Ceramic particle reinforced brazing alloy

One strategy of ceramic particle reinforcement is to introduce a stable particle reinforcement phase. To ensure that the particles can be maintained in the metallurgical reaction with active elements, the particle size is usually selected as a micron or ceramic phase containing active elements is used. Qin [36] selected SiC particles with an average size of 4.6 μm to reinforce AgCuTi for brazing the C/C composite with TC4 alloy, As shown in Fig. 2(a), 15vol% SiC was distributed in the joint, which formed a reaction interface with the Ti element in the brazing alloy. It ensured the effective binding between the brazing alloy and the particle, and the joint strength increased from 22MPa to 29MPa. Wang [37] selected TiC particle reinforced AgTi filler metal to braze C/C composite material and GH3044 alloy. TiC could be effectively combined with Ti without consumption. As shown in Fig. 2(b), when the TiC content increased to 24vol%, the joint strength reached from 40MPa to 67.2MPa. According to the finite element calculation, the residual stress was reduced by nearly 20.1%. It is worth noting that the distribution uniformity of micron-scale reinforced particles is limited, and the enhancement effect has a gap with the nanoscale phase. According to the interface reaction between Al_2O_3 and AgCuTi [16], they are connected by the Ti—O reaction layer and will not dissolve Al_2O_3 after the formation of a stable interface. Accordingly, Zhou [38] used nano-sized Al_2O_3 particles reinforced AgCuTi filler metal brazing C/C and TC4. As shown in Fig. 2(c), Al_2O_3 particles were uniformly distributed in the joint and could exist stably in the joint. The joint strength could reach to 27.8MPa when the addition amount was 0.3wt%. Based on the same strategy, both TiN [39] particle and B_4C [40] particle reinforced brazing alloy could achieve a good stress relief effect.



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Fig. 2. Microstructure of the ceramic-alloy joint brazed with particles reinforced brazing alloy. (a) SiC particles reinforced AgCuTi [36]. (b) TiC particles reinforced AgCuTi [37]. (c) Al_2O_3 particles reinforced AgCuTi [38]. (d) BN particles reinforced AgCuTi [41].

To further ensure the dispersion of the particles, another improved strategy for ceramic particle reinforcement is in-situ synthesis of reinforced phases. By adding ceramic phase and active elements in the brazing alloy fully, it can react to form ceramic phase products

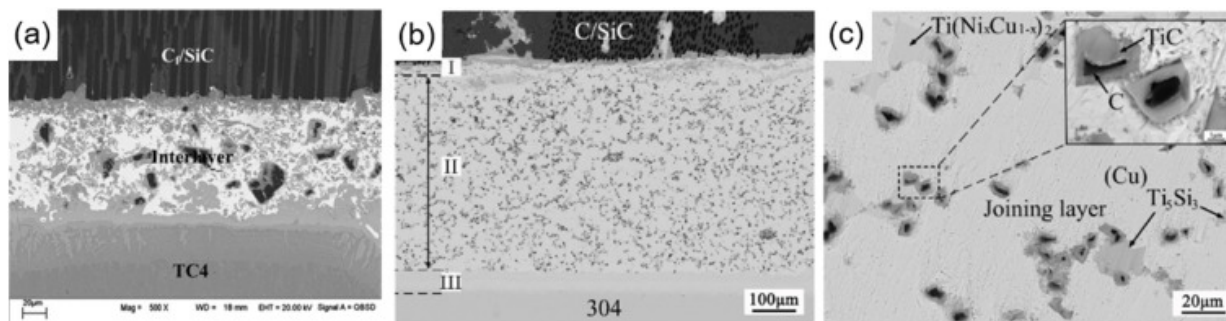
with uniform distribution. This method is often used for brazing alloys with high active element content, or in combination with additional supplemental active elements. Yang [41] mixed 3wt% BN particles in AgCuTi brazing alloy to relieve residual stress when brazing SiO₂-BN ceramics and pure titanium. As shown in Fig. 2(d), titanium alloy fully diffused and dissolved into the joint, and BN particles were completely reacted into the TiN phase and TiB whisker. Both of them were distributed diffusively throughout the joint to fully relieve the residual stress of the joint. The joint strength was increased to 31.4MPa, which was 3.4 times higher than that of the joint without the reinforced phase. The in-situ synthesis method not only distributes the reinforced phase uniformly but also consumes and controls the active elements dissolved by the metal material into the brazing alloy, avoiding the excessive reaction with the ceramic material. In the reaction with active elements, Si₃N₄ [42], B [43], TiO₂ [44] and WC [45] particles can all be transformed into a uniformly distributed strengthening phase, which ensures the stability and uniformity of joint properties. The in-situ synthesis method has good dispersion and reinforced phase size, but the coefficient of thermal expansion of the resulting compounds is much higher than that of traditional ceramics. For example, a titanium-based compound has a coefficient of thermal expansion of about $7 \times 10^{-6} \text{K}^{-1}$, The coefficient of thermal expansion of traditional ceramics such as SiC, and Al₂O₃ is generally lower than $4 \times 10^{-6} \text{K}^{-1}$, At the same addition amount, the reduction of the coefficient of thermal expansion of traditional ceramics is more obvious, which also means a better stress relief effect.

In general, most ceramic particles have good stability in the joint, and they have a lower coefficient of thermal expansion and usually have better effects. It is also the most widely used in particle reinforced methods. The in-situ synthesis of ceramics also allows for moderate consumption of reactive elements, and alleviates overreaction at the interface. Avoiding agglomeration and ensuring a uniform distribution of particle phases can better ensure the effective performance of this property.

3.2. Carbon reinforced brazing alloy

Carbon matrix materials also have a very wide range of applications in the relief of residual stresses. Carbon materials have a lower thermal expansion coefficient and a more desirable residual stress regulation effect. Due to the different arrangement of atoms, carbon materials have a very wide variety of species. Different carbon materials also exhibit different properties, and thus they have different characteristics for relieving residual stresses.

The primary application of carbon material in the brazing field is adding graphite powder to brazing alloy. Cui [46] added 7 μm carbon to AgCuTi, as shown in Fig. 3(a), larger carbon particles were connected to the filler metal by the TiC reaction layer, The strength of the C/SiC-TC4 joint could reach from 102MPa to 145MPa. Short carbon fibers as reinforced phases also have a similar interface structure [47]. Wang [48] chose 3–5 μm carbon powder to strengthen CuTi brazing alloy. At higher brazing temperature and longer holding time, most of the small-size carbon particles were converted into TiC phase, and would have a better distribution state. As shown in Fig. 3(b) and (c), the carbon content could reach 10 at.% of the brazing alloy. The joint strength of the C/SiC-304 steel joint could reach 177.8MPa, which was increased by 55.6%, showing a good stress relief effect.

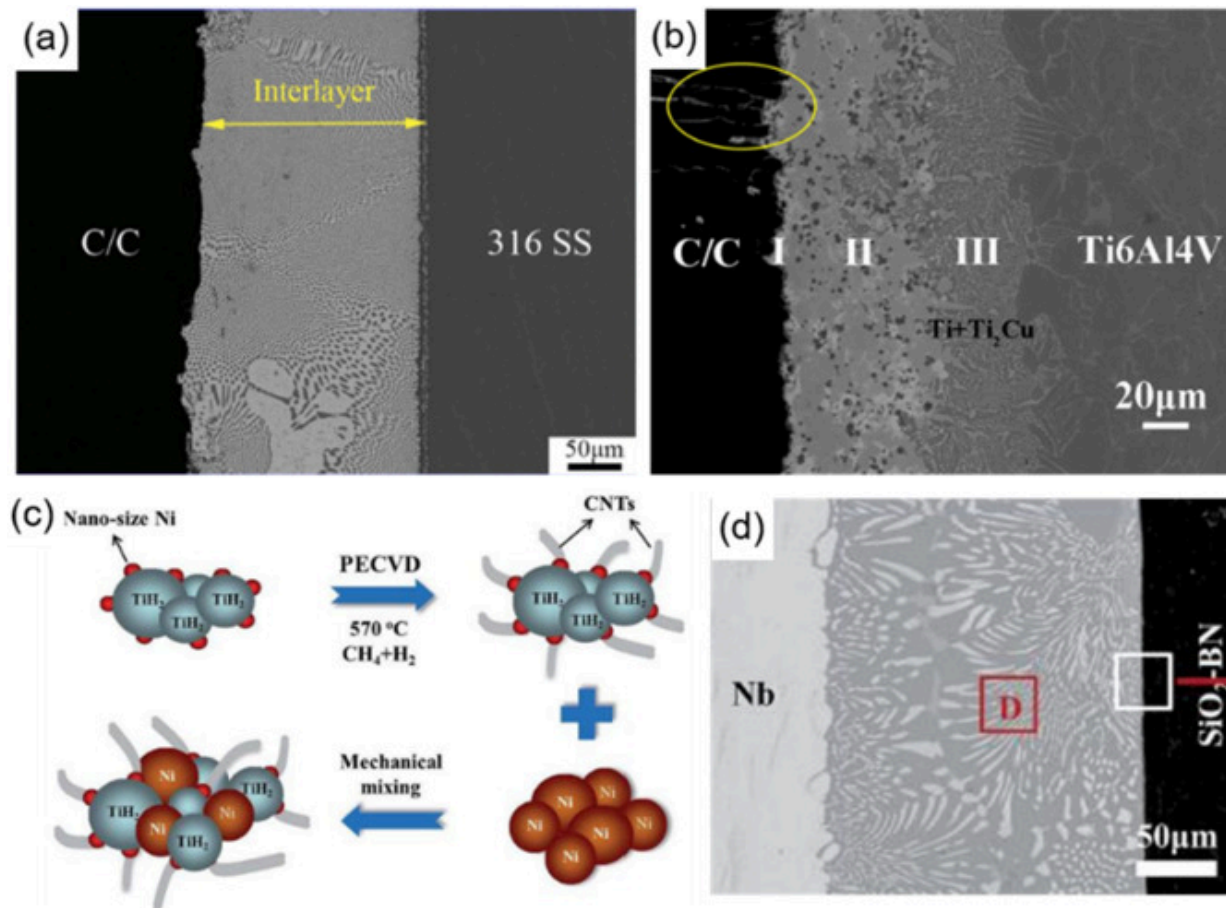


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Fig. 3. The effect of carbon fiber size on the microstructure of brazing joint. (a) AgCuTi reinforced with carbon particles of 7 μm [46]. (b) CuTi reinforced with carbon particles of 3–5 μm [48] and (c) the magnified image of (b) [48].

The improvement of metal properties by nanostructured new carbon materials has been fully demonstrated in recent years. To further reduce the reinforced phase size and explore the mitigation of residual stress by different carbon materials, Zhang [49] fully mixed 15vol% diamond particles with AgCuTi by ball milling. As shown in Fig. 4(a), the ball milling process fully refined the diamond size and evenly distributed the diamond. The Ti—Cu phase was refined and discontinuous, and the formed TiC was difficult to observe directly. The joint strength of C/C-316 steel reached 53MPa, which was nearly doubled. In view of the excellent performance of nanoscale carbon materials, Song [17] fully mixed TiZrNiCu with 1wt% carbon nanotubes by ball milling, and the strength of the joint between C/C and Ti6Al4V was increased by nearly 74%. In Fig. 4(b), it could be observed that the black TiC phase was distributed in the joint, indicating that the CNTs were agglomerated during the mixing process, which was not conducive to grain refinement and property uniformity. This agglomeration phenomenon also occurs sometimes when graphene nanosheets are selected as the reinforcing phase [50]. Agglomeration limits the increase of the amount of reinforced phase and the improvement effect. Uniform distribution of nanophase is the key to its excellent performance. To solve this problem, Qi [51] took advantage of the extremely high specific surface area of brazing alloy powder and in situ synthesized carbon nanotubes on its surface. As shown in Fig. 4(c), the content of carbon nanotubes in TiNi filler metal could be increased to 1.5vol%, and the brazing alloy could directly braze SiO₂-BN and Nb without mixing. The interface structure was shown in Fig. 4(d). All phases were uniformly distributed in the brazing alloy and reinforced phase was not observed. The joint strength was increased to 85MPa, achieving an increase of 73%. In conclusion, ensuring a good fit and uniform distribution between the reinforced phase and the brazing alloy was the basis to relieve stress and strengthen.



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Fig. 4. Microstructure of brazing joint brazed with nano-scale carbon materials reinforced brazing alloy. (a) Diamond reinforced AgCuTi mixed by ball-milling [49]. (b) Carbon nanotubes reinforced TiZrNiCu mixed by ball-milling [17]. (c) Schematic diagram and (d) microstructure of brazing alloy reinforced with in-situ synthesized carbon nanotubes [51].

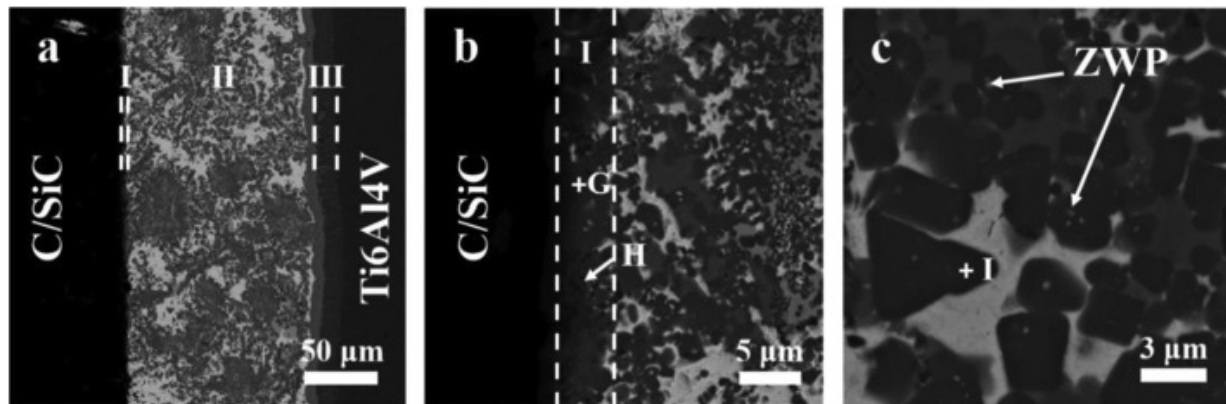
Based on above analysis, ensuring good coordination between the reinforcing phase and the brazing material, as well as a uniform distribution, is the basis for the stress relief and

strengthening effect of the reinforcing phase. The carbon materials can either be distributed in the joint to blend the stress or combined with the active brazing material to form ceramic particles that consume the active elements to also relieve the residual stress. Keeping the carbon material uniformly distributed in the joint to blend the stress and avoid agglomeration is the key to its next research.

3.3. Negative expansion material reinforced brazing alloy

Through the study of many reinforced phases, the dispersion of the reinforcing phase and the compatibility with the brazing alloy is solved. To fully reduce the property difference, particle addition is still the bottleneck problem limiting such methods. Joints with more than ideal content will lead to joint defects such as agglomeration and holes. According to the mixture theory [formula \(1\)](#), the development trend of particle reinforced composite brazing alloy is to select a lower coefficient of thermal expansion strengthening phase and fully increase its content. Recent studies have demonstrated zero-expansion metal matrix composites using negative expansion particles [\[52\]](#), [\[53\]](#). If it is applied to the brazing field, it will get a more excellent property mitigation effect than carbon material. As shown in [Fig. 5](#), Ba [\[54\]](#) pioneered to enhance AgCu brazing alloy with nano-sized negative expansion particles $\text{ZrP}_2\text{WO}_{12}$ ($\text{CTE} = -2 \times 10^{-6} \text{K}^{-1}$). And the finite element analysis showed that the residual stress was reduced by 52.9MPa and the joint strength was increased to 146.2MPa (an increase of 70.8%) by adding only 3wt%. The particles formed a stable interface with Ti and were not fully consumed, which could ensure the development of its negative expansion characteristics. In addition to $\text{ZrP}_2\text{WO}_{12}$ particles, $\text{Y}_2\text{W}_3\text{O}_{12}$ ($\text{CTE} = -12 \times 10^{-6} \text{K}^{-1}$) [\[55\]](#) and $\text{Sc}_2(\text{WO}_4)_3$ [\[56\]](#), [\[57\]](#) ($\text{CTE} = -5.6 \times 10^{-6} \text{K}^{-1}$) also had a good stress relief effect as negative expansion materials. The negative expansion property was maintained during brazing. The negative expansion particle enhancement method also played an ideal effect of regulating the coefficient of thermal expansion in glass filler metal and air reaction filler metal [\[58\]](#), [\[59\]](#). For example, Si [\[59\]](#) added 6wt% lithium sauwnet ($\beta\text{-LiAlSiO}_4$) to the air reactive brazing alloy Ag-CuO, and the coefficient of thermal expansion of the filler alloy decreased from $19.6 \times 10^{-6} \text{K}^{-1}$ to $12.2 \times 10^{-6} \text{K}^{-1}$. And the shear strength of the joint reached from 45MPa with the content of 2wt% to 62MPa

with the content of 6wt%. When Qi [60] applied lithium nepheline to AgCuTi brazing alloy, it was found that when the content of lithium nepheline exceeded 5vol%, a large number of holes and unwelded defects would occur. This was due to the significant depletion of active elements and the very poor wettability with oxides. The strength of C/C and Nb was increased by nearly 2.3 times by preparing lithium nepheline/Cu+composite interlayer.



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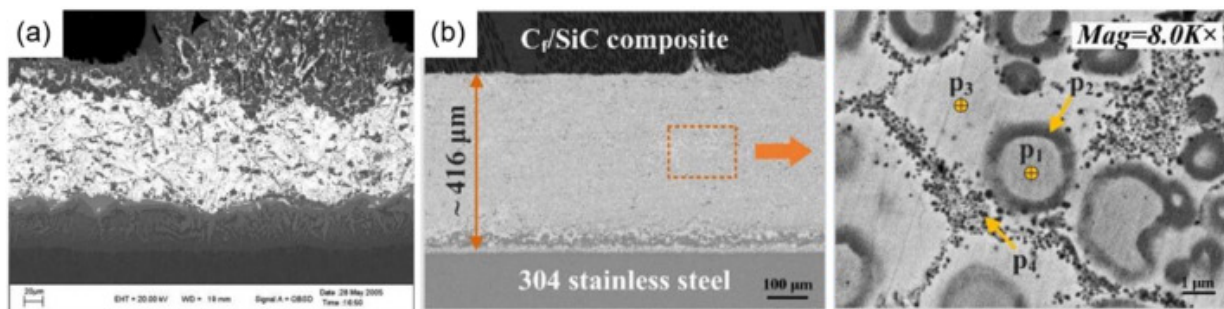
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Fig. 5. SEM images of C/SiC-Ti6Al4V joints brazed with AgCu+3wt% ZWP [54]. (a) the whole joints, (b) reaction layer at C/SiC side and (c) brazing seam.

In general, the negative expansion material can compensate for the mismatch of thermal expansion of the joint. But it can also introduce unwanted reactions into the joint, which can cause certain defects. Therefore, the research direction of particle reinforced negative expansion material should maintain the negative expansion property of the material without introducing complex reactions in the joint. Negative expansion materials in combination with other methods to synthetically relieve residual stresses will be a future research direction.

3.4. Metal particle reinforced brazing alloy

One of the important reasons for the limited number of ceramic particles is that the interfacial reaction between ceramic and brazing alloy consumes a large number of active elements. If the metal particles with low expansion are used, the interfacial bonding can be completed by solution and diffusion with the filler alloy. When Lin [61] brazed C/SiC and titanium alloy by AgCuTi, the addition amount of W particles ($4.5 \times 10^{-6} \text{K}^{-1}$) could be as high as 50vol%. As shown in Fig. 6(a), the joint was basically filled with W particles. The residual stress was greatly relieved, and the joint strength could be increased to 180MPa. The introduction of a large number of W particles was bound to improve the liquidus and viscosity of the brazing alloy, which required higher brazing temperature and longer holding time to ensure the integrity of the joint. A good regulation effect could also be achieved by introducing a large number of W particles into TiZrNiCu filler alloy with high active element content [62]. Wang [63] used another low-expansion Mo particle ($5 \times 10^{-6} \text{K}^{-1}$) to strengthen AgTi brazing alloy. As shown in Fig. 6(b), 10 at.% Mo powder could be evenly distributed in the joint. The interfacial bonding with a small amount of Ti element in solid solution was effective in relieving the stress of C/SiC-Nb joint, and the joint strength could reach 156MPa.



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Fig. 6. Microstructure of the joint brazed with low CTE alloy reinforced brazing alloy. (a) W particles reinforced AgCuTi alloy [61]. (b) Mo particles reinforced AgTi alloy [63].

It can be found that metals with low expansion properties like W and Mo can also play a role in regulating the residual stress in the joint. Since it is also a metal material with similar properties as the brazing material, the bonding between the brazing material and the modulating metal is mainly through solid solution diffusion. However, it can also have an impact on the overall process parameters of the brazing material. In general, the reinforced phase particles have a very good effect on the residual stress relief of the joint. The researches mentioned above about the particle reinforced brazing alloy methods are concluded in [Table 1](#) to show the residual stress relief effects straightforwardly. However, there is still a need for research on issues such as agglomeration and active brazing material consumption. The method of particle-reinforced phase brazing with a variety of regulation methods will be a focus of future research.

Table 1. The residual stress relief effects of particle reinforced brazing alloy methods.

Type of particles		Authors	Relief effect
Ceramic particle	SiC	Qin [36]	From 22MPa to 29MPa
	TiC	Wang [37]	The residual stress reduced by nearly 20.1%
	Al ₂ O ₃	Zhou [38]	Reached to 27.8MPa, higher than other researches
	B ₄ C	Li [40]	From 62MPa to 98MPa
	BN	Yang [41]	Increased to 31.4MPa, 3.4 times higher than that of the joint without the reinforced phase
	Si ₃ N ₄	Song [42]	From 53MPa to 115MPa
	B	Yang [43]	The shear strength increased by 23.5%
	TiO ₂	Zang [44]	The residual stress reduced by 30%
	WC	He [45]	From 114.5MPa to 320.5MPa

Type of particles		Authors	Relief effect
Carbon materials	Graphite powder	Cui [46]	From 102MPa to 145MPa
	Short carbon fibers	Wang [48]	The shear strength increased by 55.6%
	Diamond particles	Zhang [49]	The shear strength reached to 53MPa, which was nearly doubled
	Carbon nanotubes	Song [17] Qi [51]	The shear strength increased by 74% The joint strength was increased to 85MPa, achieving an increase of 73%
Negative expansion material	ZrP ₂ WO ₁₂	Ba [54]	The joint strength was increased to 146.2MPa with an increase of 70.8%
	Y ₂ W ₃ O ₁₂	Wang [55]	Increased to 42MPa, 1.6 times higher than that of the joint without the reinforced phase
	Sc ₂ (WO ₄) ₃	Wang [56]	Increased to 64MPa, 2.6 times higher than that of the joint without the reinforced phase
		Wang [57]	Increased to 77MPa, 3.2 times higher than that of the joint without the reinforced phase
	KAlSiO ₄	Lin [58]	The shear strength increased by 109.6%
	β-LialSiO ₄	Si [59] Qi [60]	From 45MPa with the content of 2wt% to 62MPa with the content of 6wt%. The joint strength was increased by nearly 2.3 times
Metal particle	W	Lin [61]	From 102MPa to 180MPa
		Cui [62]	From 120MPa to 166MPa

Type of particles	Authors	Relief effect
Mo	Wang [63]	156MPa at room temperature

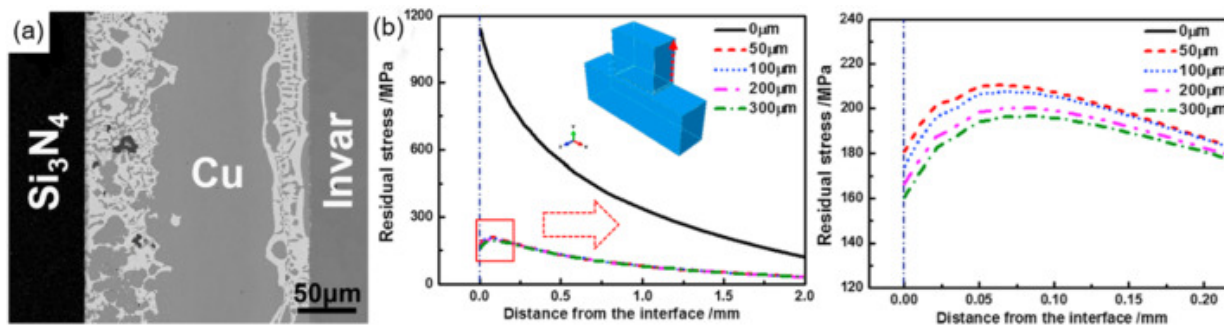
4. Interlayer assisted control residual stresses

During the cooling process of the brazed joint, residual stress will be generated if the material shrinkage exceeds the joint structure limit. The joints with high strength and low plasticity will produce extremely high stress and even fracture. To improve the plastic deformation of joints and relieve the stress, it is an effective method to regulate the residual stress through the interlayer. Usually, metal and alloy foil with good plasticity are used as the interlayer. The joint is brazed by a “sandwich” structure sandwiched between two filler brazing alloys. The sandwich interlayer is often prepared by rolling or resistance welding in the industry.

4.1. Metal interlayer

According to the selection and collocation of common brazing metal, Cu foil, Ni foil and Mo foil with good plasticity are often used as the interlayer to assist brazing [\[64\]](#), [\[65\]](#), [\[66\]](#). Wang [\[67\]](#) studied the influence of Cu interlayer thickness on the interface structure and residual stress of joints. As shown in [Fig. 7\(a\)](#), Si_3N_4 and Invar were brazed with AgCuTi sandwiched Cu interlayer. The surface layer of the Cu interlayer was dissolved in the brazing alloy to realize the connection and maintain the basic thickness. The influence of plasticity improvement on residual stress was shown in [Fig. 7\(b\)](#). When Cu was introduced, the residual stress was significantly reduced. The shrinkage limitation was offset by deformation, and the residual stress decreased gradually with the increase of Cu interlayer thickness. The shear strength of the joint increased gradually with the increase of the thickness of the interlayer, and the strength of the joint could reach 256MPa when the thickness of the middle layer is 200 μm . It is worth noting that the increase of the

thickness of the interlayer has a great influence on the size structure and stress of the joint. In this study, when the thickness of the interlayer reached 300 μm , the joint strength decreased sharply. Therefore, the thickness of the interlayer should be reasonably selected and the brazing process should be adjusted accordingly. In addition, the interlayer can introduce a high volume of reinforcement on a smaller specific surface. Based on this, some researches introduced a low expansion interlayer into the joint to relief the residual stress. W interlayer [68] and Al_2O_3 interlayer [69] were introduced into the brazed joint and achieved certain results. However, the problem was also more prominent, both the ceramic and the interlayer were not compatible with the properties of the brazing alloy between them. Regional stress regulation still face challenges, the distribution form of the interlayer in the joint is the key problem of interlayer strengthening.



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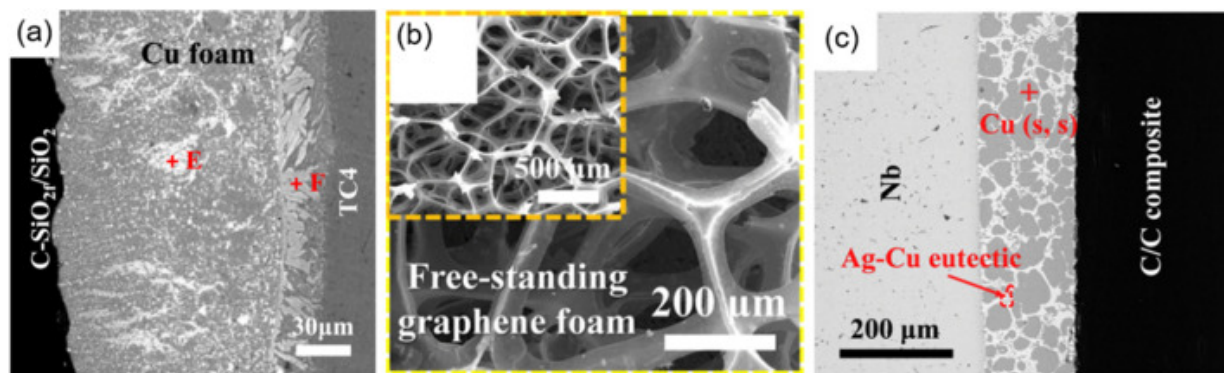
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Fig. 7. Microstructure and residual stress analysis of the Si_3N_4 -Invar brazing joint modified with Cu interlayer [67]. (a) Microstructure of the brazing seam modified with Cu interlayer. (b) Effect of the Cu thickness on the residual stress.

4.2. Porous interlayer

To change the stratified distribution of the interlayer, and improve the distribution of the interlayer in brazing, the porous structure becomes the optimal form of the interlayer.

Porous metals are widely used in the stress regulation of brazing structures. Many studies have analyzed the of porous metal foams such as Cu [70], Ni [71], Fe [72] and polyurethane [75] on the interface structure and residual stress of joints. The porous interlayer is distributed uniformly throughout the joint in 3D form, which can regulate the joint plasticity as a whole. Since the ratio of length to width of porous material is high, the structure of porous material is easy to dissolve during brazing. When using porous Cu foam to assist brazing $\text{SiO}_{2f}/\text{SiO}_2$ -TC4 joint, Lin [70] found that the porous Cu foam was fully dissolved at the brazing temperature of 900°C. As shown in Fig. 8(a), the porous Cu foam dissolved into the AgCu matrix and reacted with the high content of Ti element in the joint, forming a large number of diffused Ti—Cu phases. To prevent the dissolution and collapse trend of Cu foam, Wang [73] coated a layer of graphene on the surface of Cu foam in situ by CVD technology. As shown in Fig. 8(b), dense graphene was used to hinder the diffusion and reaction between porous Cu and brazing alloy. When using it to braze C/C-Nb joint, as shown in Fig. 8(c), the porous Cu only deformed but did not collapse. The stress relief effect brought by the high deformation capacity was very obvious, thus the shear strength of the joint was increased to 43MPa, which was nearly two times higher.



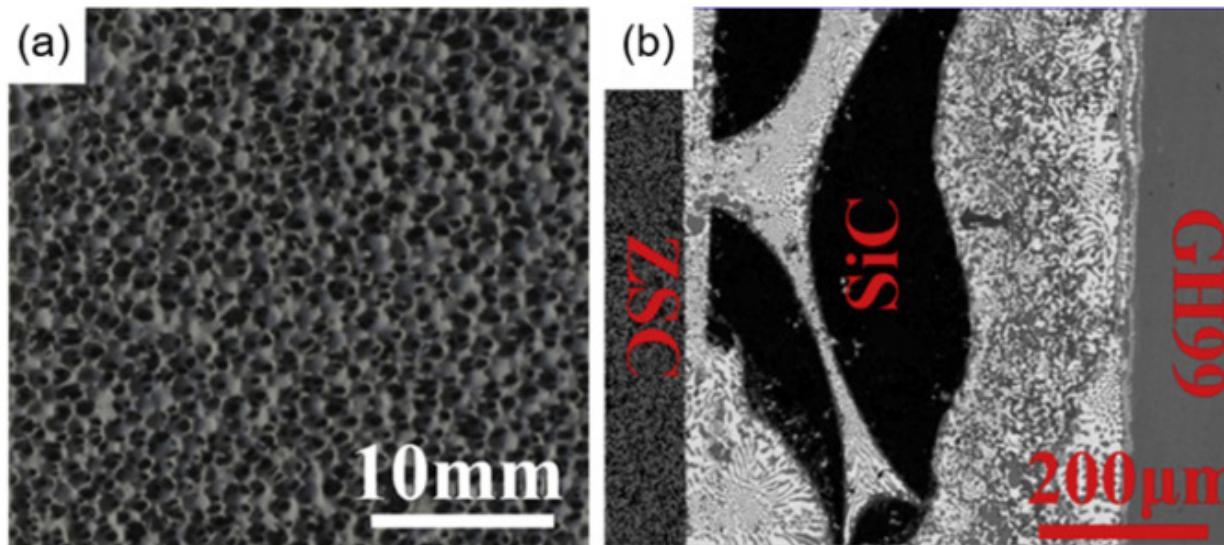
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Fig. 8. Microstructure of the brazing joint reinforced with the porous Cu. (a) Microstructure of the $\text{SiO}_{2f}/\text{SiO}_2$ -TC4 joint reinforced with Cu foam [70]. (b) Morphology

of the graphene reinforced Cu foam [73]. (c) Microstructure of the C/C-Nb joint reinforced with graphene-Cu foam [73].

The aforementioned porous metals presented dissolution and retention phenomena during the brazing process, providing two strategies for its application. Firstly, stable porous material is used to adjust the performance of the joint. The selected materials are mainly ceramic phase. In the case of much lower than the particle-specific surface area, the amount of ceramic phase added is greatly improved, and the distribution uniformity is guaranteed. Zhang [74] used porous SiC interlayer to adjust the joint when brazing the ZrB_2 -SiC-C and GH99, as shown in Fig. 9. The SiC content could be increased up to 25 vol%, and the interface between SiC and brazing alloy was intact without defects. The difference of CTE between brazing alloy and ceramic material was alleviated, and the joint strength was increased by nearly 1.6 times. Porous Si_3N_4 interlayers [41] and lithium chalcopyrite porous interlayers [42] have been studied in brazing, all reflecting the effective relief of residual stresses by high content reinforcements. The porous homogeneous structure avoids uneven properties and stress concentration and ensures the stabilizing effect of the interlayer in the joint.

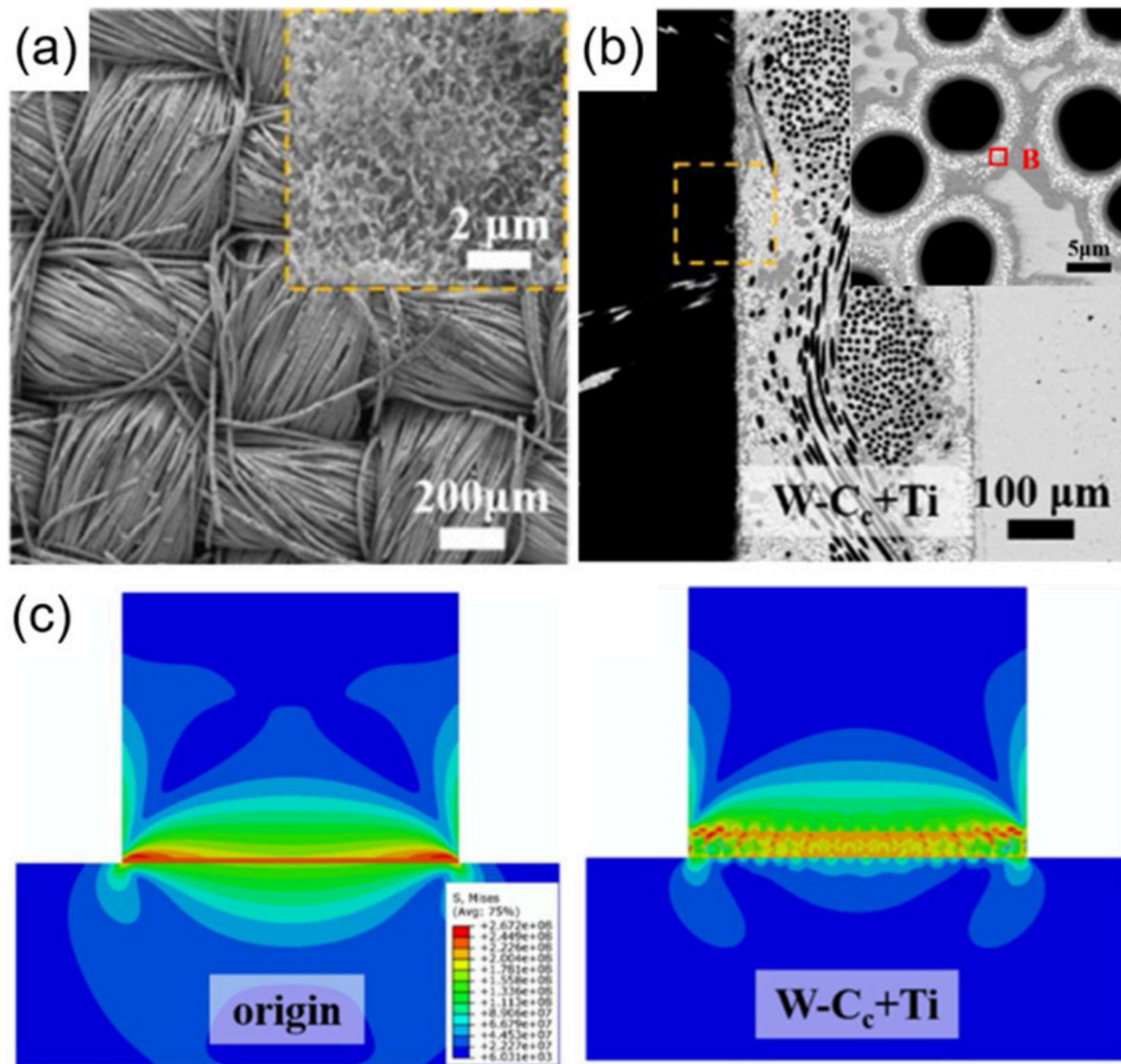


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Fig. 9. The brazing joint of ZrB_2 -SiC-C and GH99 reinforced with SiC network [74]. (a) Morphology of the SiC network. (b) Microstructure of the brazing joint reinforced with SiC network.

Another strategy for porous materials is to use a porous interlayer as a carrier. This approach allows loading the reinforced phase on the porous material and introducing the reinforced phase uniformly into the joint through the porous material. For example, when carbon nanotubes are grown in situ on the surface of porous Ni, the carbon nanotubes on the surface of porous Ni are uniformly distributed in the joint. The content of carbon nanotubes could be increased to 2.7 vol%, which could effectively reduce the residual stress and refine the joint grain structure [76]. Similar results were obtained by in situ synthesis of CuO nanosheets on the surface of porous Cu [77]. By using stable carbon fiber cloth as the porous interlayer and synthesizing nano-tungsten phase on its surface in situ, the advantages of the two can be combined simultaneously, as shown in Fig. 10(a) [33]. In the brazing process, as shown in Fig. 10(b), under the reaction drive of Ti and carbon fiber, the brazing alloy formed a TiC reaction layer on the surface of carbon fiber to realize bonding. Weakly bound tungsten nanoparticles were diffusely distributed near the fiber surface, to reduce the properties difference between the fiber and the brazing, and the residual stress caused by the interlayer. From the finite element analysis, it could be obviously found that the introduction of low expansion fiber cloth could fully reduce the state of high residual stress in the joint. Under the control of tungsten nanoparticles, no obvious high stress was generated between the fiber and the filler metal.



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Fig. 10. Nano tungsten reinforced the carbon fiber cloth for brazing C/SiC and Nb [33]. (a) Morphology of the nano tungsten reinforced the carbon fiber cloth. (b) Microstructure of

the brazing joint reinforced with the nano tungsten reinforced the carbon fiber cloth. (c)

Effect of the nano tungsten reinforced the carbon fiber cloth on the residual stress.

On the one hand, the application of porous interlayer improves the distribution state of the interlayer in the joint, and provides more material selection and content improvement. Its large specific surface area can be used as the carrier of particle reinforcement phase to evenly distribute particles to prevent agglomeration. On the other hand, the combination of particles and porous interlayer can not only expand the application environment of the particle phase, but also make up for the regulation reaction of the large-size of porous materials. Various forms of collocation will provide the most ideal design scheme for specific brazing examples. Further, the residual stress relief effects of interlayer assisted control methods were concluded in [Table 2](#).

Table 2. The residual stress relief effects of interlayer assisted control residual stresses methods.

Type of interlayer		Authors	Relief effect
Metal interlayer	Cu foil	Zhao [64]	The residual stress reduced from 384MPa to 119MPa
		Wang [67]	The joint strength was increased to 256MPa with an increase of 47.9%
	Ni foil	Feng [65]	The joint strength reached to 110MPa
	Mo foil	Li [66]	From 5.3MPa to 98.5MPa
	W foil	Sun [68]	Increased to 77MPa, 175% higher than that of the joint without the interlayer
Porous metal interlayer	Porous Cu	Lin [70]	From 7.5MPa to 59.6MPa
	Porous Ni	Zhu [71]	From 12.56MPa to 119.86MPa

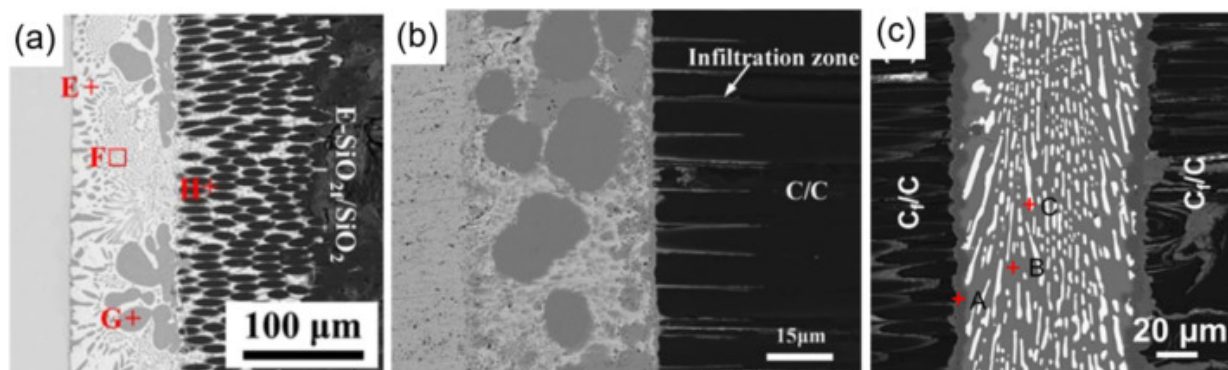
Type of interlayer		Authors	Relief effect
Porous ceramic interlayer	Porous Fe	Shirzadi [72]	From 22 MPa to 33 MPa
	Porous SiC	Zhang [74]	Increased by nearly 1.6 times
	Porous Si ₃ N ₄	Yang [41]	Increased by 161%
Porous interlayer+reinforced phase	Porous lithium chalcopyrite	Song [42]	Increased to 115 MPa, 85% higher than that of the joint without the interlayer
	Porous Cu+graphene	Wang [73]	Increased to 43 MPa, 2 times higher than that of the joint without the interlayer
	Porous Ni+carbon nanotubes	Ba [76]	Increased to 50 MPa, 5 times higher than that of the joint without the interlayer
	Porous Cu+CuO nanosheets	Wang [77]	Increased to 95.6 MPa, 83.1% higher higher than that of the joint without the interlayer
	Porous carbon fiber+W	Xie [33]	Increased to 131.8 MPa, 57% higher higher than that of the joint without the interlayer

5. Surface structure design

The properties of ceramics and metals change at the interface, and high stress is concentrated there. Because the ceramic side reaction layer is mostly of brittle compounds, it becomes the weak part of the joint, and fracture often occurs in this part.

By designing the surface structure of ceramic or its composite material, the surface structure of ceramics or their composites is designed to change the planar reaction layer into a curved or more complex structure. This not only increases the connection area and enhances the pinning effect, but also enables a gradient transition of properties through the interlocking interface between ceramics and metals. Mechanical drilling [78] and laser drilling [69], [79], [80], [87] can effectively design ceramic surface structure. Hernandez [78] performed mechanical drilling on C/SiC surfaces and arrayed drilling on large-size C/SiC plates, when C_f/SiC and Ti6Al4V alloys were brazed with AgCuTi, the holes were fully filled with brazing alloy and the pinning effect was formed. The results showed that the joint strength increased nearly three times and even exceeded the strength of C/SiC even when the strength of C/SiC was very low. Similarly, Wang [86] brazed C/C composites and TiAl alloy with AgCuTi alloy, holes were manufactured by mechanical drilling, and the transition structure reduced property differences relieving the residual stress effectively, the shear strength of the joint increased by 137% compared with the joint without drilling. Further, Shen [69] used laser drilling to process a series of holes 300–1000μm deep on the surface of C/C composites when brazing C/C composites with Ni-based high-temperature alloys with AgCu brazing material, typical pinning effect was formed with the brazing alloy penetrating into the C/C composites, the shear strength enhanced from 16MPa to 73MPa. Likewise, Li [79] introduced laser drilling to YSZ surface to braze YSZ and TC4, the residual stress distribution in the joint becomes non-linear and the cracks extension was prevented, the shear strength of the joint increased to 150MPa, 95.2% higher than that of the joint without drilling. Tian [80] performed laser drilling on the surface of C/SiC, brazed C/SiC and Nb with TiNi brazing alloy. Since the filler alloy penetrated into the hole of the ceramic and formed a pinning structure, the filler alloy bears the load and stress of the ceramic, the fracture path was prolonged, the residual stress of the joint was reduced and dispersed, and the final shear strength reached 259MPa. Above all, Mechanical and laser processing is subject to mechanical cutting forces and high heat input, which can be detrimental to the strength and denseness of ceramic materials. The stresses caused by the large-sizes machined are also a major problem.

To further reduce the pinning size and ceramic damage, the fiber in fiber-reinforced ceramic matrix composites is used to form a gradient transition structure with the filler metal. Since the degree of transverse and longitudinal corrosion of fiber materials can be different, the corrosion surface modification of different fiber materials can be achieved by various corrosion methods [81], [82], [83]. This can form a gradient composite structure to achieve joint reinforcement. The high-frequency corrosion treatment of $\text{SiO}_2/\text{SiO}_2$ by Ma [81] was shown in Fig. 11(a). After etching, the wettability of ceramic materials was greatly improved. The brazing alloy penetrated into the ceramic material to form a gradient composite structure, which reduced the residual stress of the joint, and the shear strength of the joint reached 61.9MPa. Besides, Ba [82] adopted thermal corrosion to treated SiC_f/SiC surface, the SiC fibers were exposed to thermal corrosion at low temperatures to form a SiC fiber-reinforced AgCuTi transition layer, the crack expansion was hindered and the shear strength of the joint reached to 41.4MPa with an increase of 20.7MPa. Another corrosion by Ba [83] was to use electrochemical corrosion at the C/C composites surface. C fiber in C/C composites surface was corroded in different depth, the shear strength reached to 37.7MPa with 1.2 times higher than that of original joint. As described, the corrosion surface modification mainly applied to composites materials, reinforced by the pinning of fibers.



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Fig. 11. The design of surface structure in the fiber reinforced brazing alloy. (a) Microstructure of brazing Nb and $\text{SiO}_{2f}/\text{SiO}_2$ modified with the chemical corrosion [81]. (b) Microstructure of brazing Nb and C/C modified with the oxidation [84]. (c) Microstructure of brazing C/C with $\text{Si}-\text{Zr}$ brazing filler [85].

Further, there were different surface design methods for different materials. For instance, Yang [84] preoxidized C/C composite ceramics as shown in Fig. 11(b). Through preoxidation, adjustable annular gaps were formed around the carbon fiber. These gaps allowed the brazing alloy to penetrate into the composite ceramic, and finally increased the shear strength of the joint from 29MPa to 57MPa. Moreover, when He [85] brazed C/C composite materials, He chose $\text{Si}-\text{Zr}$ filler alloy with high Si content, and used the reaction between liquid Si and C to realize the connection. As the reaction between Si and C went deeper and deeper, Si gradually diffused into C/C to form SiC phase, the fiber could be retained to realize the fastening of the brazing alloy as shown in Fig. 11(c). If the joint was brazed with the composite filler alloy, the joint quality would be improved as before. Similarly, $\text{Ti}-\text{Si}$ alloy was used in the same way to braise C/C composites, the $\text{Ti}-\text{Si}$ eutectic structure also exhibited a gradient structure. This method was more effective than direct brazing due to the strong interfacial bonding and gradient structure.

To summarize, surface structure design mostly focuses on the stress concentrations in the joint which consists in the interface between ceramic or ceramic matrix composites and brazing alloy. Stress relief in the form of gradient transition would benefit the mechanical properties of the joint. Finally, the residual stress relief effects of surface structure design methods were concluded in Table 3.

Table 3. The residual stress relief effects of surface structure design methods.

Type of surface structure design		Authors	Relief effect
Drilling	Mechanical drilling	Hernandez [78]	The residual stress reduced from 384MPa to 119MPa

Type of surface structure design		Authors	Relief effect
Corrosion	Laser drilling	Wang [86]	Increased by 137%
		Shen [69]	From 16MPa to 73MPa
		Li [79]	Increased to 150MPa, 95.2% higher than that of the joint without drilling
		Tian [80]	From 192MPa to 259MPa
	High-frequency corrosion	Guo [87]	From 18.7MPa to 31 MPa
		Ma [81]	From 5MPa to 61.9MPa
		Ba [82]	From 20.7MPa to 41.4MPa
Preoxidized	Thermal corrosion	Ba [83]	Increased to 37.7MPa, 1.2 times higher than that of the joint without corrosion
	Electrochemical corrosion		
	Preoxidized	Yang [84]	From 29MPa to 62MPa
High temperature penetration	Zr-Si filler alloy	He [85]	Reached to 32MPa
	Ti-Si filler alloy	He [88]	Reached to 26MPa

6. Conclusions and perspective

In response to the special needs of ceramic-metal composite components in aerospace, electronics, and instrumentation, this review focuses on the core problem of excessive residual stresses in ceramic-metal brazed connections. Due to the large difference in thermal expansion coefficients between ceramics and metals, excessive residual stresses in joints can cause failure. This review analyzed the causes of this generation and proposed a series of analysis methods for residual stresses. To effectively address this issue and adequately relieve residual stresses in the joint, this review focuses on three

types of methods to relieve residual stresses. We focused on analyzing the effects of various approaches on the structure of the brazing interface and elucidated the effects and principles of residual stress modulation. The method of particle reinforcement was to distribute the state of stress concentration by diffusely distributed particles, which could withstand the load and adjust the brazing alloy properties to regulate the stress. Interlayer-assisted regulation was to improve the overall plasticity of the joint by an interlayer between brazing alloys, which improved the overall joint deformation relief and hence regulated the overall stress. The surface structure design was to enhance the pinning effect on the ceramic side by surface modification of the ceramic side, which could create a gradient transition and relieve the stress concentration at the interface of the ceramic and brazing alloy. These methods of stress regulation are very instructive for ceramic-metal connections, and they can effectively improve the service performance of ceramic-metal composite components. In response to these residual stress relief methods, we have comprehensively evaluated future research directions for ceramic-metal brazed connections. The improvement of these methods and the application of the technology will set off a new revolution in the fields of aerospace, automotive, electronics, and energy.

Firstly, composite approach to stress regulation is urgently needed. Single stress regulation methods have their own drawbacks, but combining multiple methods can effectively exploit the advantages of different methods and avoid their drawbacks. For example, the addition of the particulate phase requires avoiding the agglomeration of the particulate phase and ensuring their dispersion. The foam interlayer needs to be considered to avoid the collapse of the three-dimensional structure. When combining the two methods, the presence of the three-dimensional structure ensures the dispersion of the particle phase, and the presence of the particle phase guarantees that the three-dimensional structure does not collapse. This guarantees that both methods can be used effectively at the same time, thus providing better relief of residual stresses. The composite approach to stress regulation will be more widely used to bring into play the regulation characteristics of different methods and avoid the defects caused by a single

method, which will enable the combination of ceramics and metals to take a new step forward. Admittedly composite approach is highly helpful in relieving residual stresses, but there are still some challenges to overcome. For instance, stress regulation of multiple methods requires more cost and has lower efficiency for engineering needs. Moreover, the design of composite approach requires the advantages of advantages and prevents the disadvantages of each method, which requires insightful awareness of various regulation methods. Further, whether the various methods can act together to regulate stress also needs further investigation. Herein, there are still many difficulties in realizing the application of composite approach.

Secondly, large-size regulation is also required for practical applications. Currently the regulation of joints in experimental processes is often performed at small sizes, however, in actual industrial production, ceramic and metal composite parts are often large-size parts. Failure of such components often affects the operation of the entire machine, so this puts a higher demand on the stress regulation of the joint. During most experiments, ceramic and metal joints are usually smaller in size. If there are large residual stresses and defects in the small size joints, large joints used in engineering are subject to more residual stresses and defects. Accordingly, larger joints require a lower overall level of residual stresses and a more uniform distribution of stresses. Stress concentrations and defects in some locations can be dangerous for the entire joint. Therefore, solving the regulation of large-size joints can better fit the needs of industrial production and is the focus of research on ceramic and metal connections.

Finally, more attention should be paid in functional modulation. The focus of traditional ceramic-metal brazed joints is often on the mechanical properties of the joint. However, since the residual stress affects the joint not only in terms of mechanical properties, including corrosion resistance, thermal shock resistance. With the development and application of semiconductors, more semiconductor-metal composite components will be used in the future, and the change of conductivity is a new direction for the future. For these areas, residual stress regulation takes into account these properties while ensuring mechanical properties. This will be the focus of future research. The multi-domain-

oriented regulation will be a major basis for the realization of a wide range of applications of ceramic-metal composite in the future.

Ceramic-metal composites have shown great potential in today's productive life. The composite regulation method for functional and large-size will set off a new round in many fields such as aviation, electronics, and energy.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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